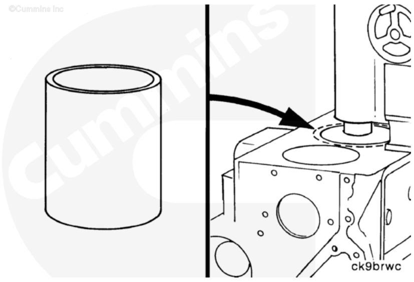
Trainings

General Information

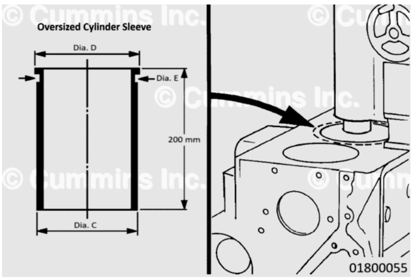


The cylinder block uses bored cylinders as opposed to liners. In the event of damage or wear out, the cylinders may be able to be repaired.

For B3.9, B4.5, and B5.9 engines, the cylinders can be bored oversize twice for the use of over size pistons and rings (0.5 mm [0.020 in]) and 1 mm [0.040 in] oversize). A repair sleeve can also be installed if the cylinder bore **must** be bored more than 1 mm [0.040 in] oversize. See the Overbore and/or Repair Sleeve section of this procedure.

For B4.5 RGT engines, the cylinders can **only** be bored oversize once for the use of oversize pistons and rings (0.5 mm [0.020 in] over size). A repair sleeve can be installed if the cylinder bore **must** be bored more than 0.5 mm [0.020 in] oversize. See the Overbore section in this procedure.

Oversized Cylinder Sleeves

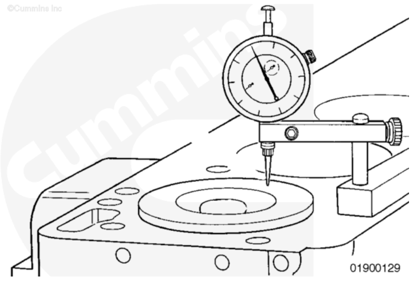


B3.9 and B5.9 engines can also use oversize repair sleeves A repair sleeve can also be installed if the cylinder bore must be bored more than 1 mm [0.040 in] oversize up to approximately 2 mm [0.080 in] See the B3.9 and B5.9 engines Oversized Repair Sleeve section of this procedure

Initial Check

Prior to removing the piston and connecting rod assemblies, measure and record piston protrusion. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-054-tr.html)

Measuring piston protrusion prior to disassembly will aid in determining if the cylinder block, if required, can be resurfaced.

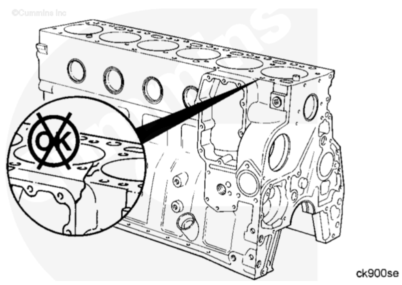


Preparatory Steps

- Remove the engine and place on an engine stand. Refer to Procedure 000-001 in Section 0. (/qs3/pubsys2/xml/en/procedures/40/40-000-001.html)
- Disassemble the engine. See Section DS - Engine Disassembly.

Initial Check:

Before cleaning or further disassembly of the block, perform a visual inspection to see if there is any damage (cracks, fretting, etc.) that would prohibit reuse. Paying close attention to areas of the block that include:



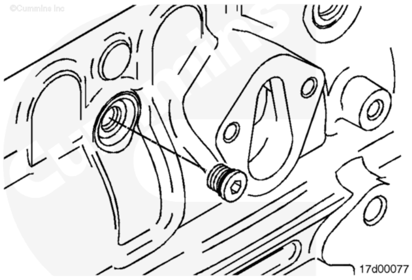
- Main bearing caps and bores
- Camshaft bores
- Cylinder bores
- Tappet bores
- Cylinder block combustion deck
- Oil pan mounting surface
- Lubricating oil pump mounting area
- Water pump mounting area
- Front and rear of block sealing surfaces
- Lubricating oil cooler cavity.

Clean and Inspect for Reuse

Inspect all pipe plugs, expansion plugs and straight thread plug for signs of damage or leaks.

If it is necessary to thoroughly clean the cylinder block for reuse due to excessive debris or contamination, remove all pipe plugs, expansion plugs and straight thread plugs as necessary. Make sure all oil and coolant passages are cleaned out.

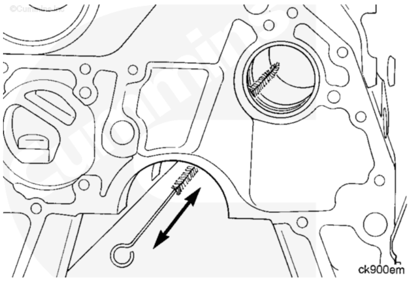
Use the following procedures for removal and installation of plugs. Refer to Procedure 017-002 (/qs3/pubsys2/xml/en/procedures/40/40-017-002.html). Refer to Procedure 017-007 (/qs3/pubsys2/xml/en/procedures/40/40-017-007.html). Refer to Procedure 017-011 (/qs3/pubsys2/xml/en/procedures/100/100-017-011.html).



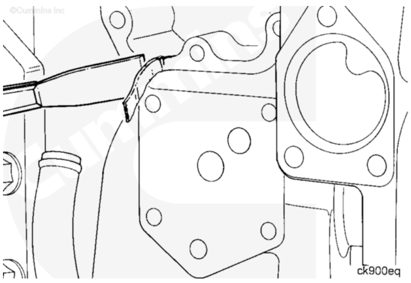
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Use clean solvent and a nonmetallic brush to clean the block oil drillings.



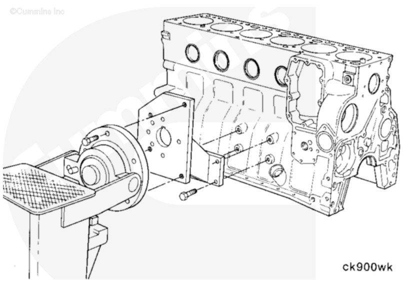
Thoroughly clean all gasket sealing surfaces of any remaining gasket residue.



⚠ WARNING ⚠

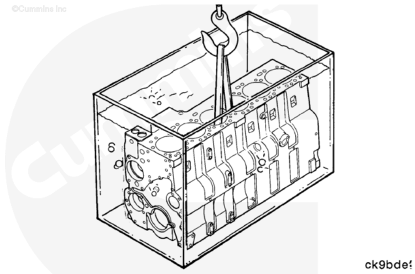
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Remove the block from the engine stand.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ CAUTION ⚠

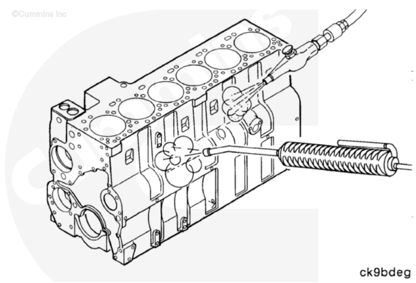
Use a cleaning solution that will not damage the camshaft bushings.

Follow the equipment manufacturer operating instructions for the cleaning tank.

Follow the solvent equipment manufacturer instructions for using the solvent.

Note : Cummins Inc. does **not** recommend any specific cleaning solution. Experience has shown that the best results are obtained by using a cleaning solution that can be heated from 80 to 95°C [176 to 203°F]. A cleaning tank that **will** mix and filter the cleaning solution will give the best results.

Clean the cylinder block in the cleaning tank.



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⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure all debris is removed from the capscrew holes and oil passages.

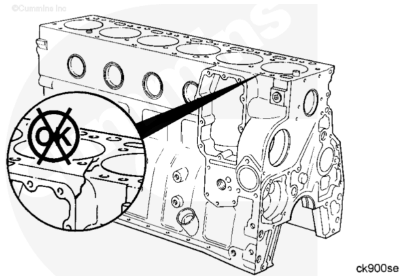
Remove the block from the cleaning tank.

Use steam to clean the cylinder block thoroughly.

Use compressed air to dry the block.

Note : If the cylinder block is not going to be used immediately, apply a coating of preservative oil to prevent rust. Cover the block to prevent dirt from sticking to the oil.

With the cylinder block cleaned, inspect the cylinder block again for signs of cracks, fretting, and discoloration that will prohibit reuse.

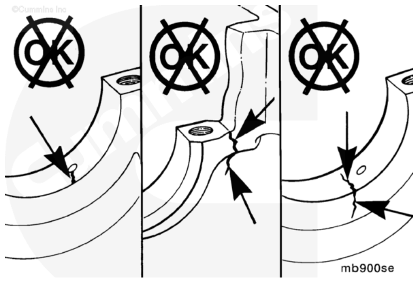


To help identify cracks in the cylinder block, use the Crack Detection Kit, Part Number 3375432.

Pay close attention to areas of the block that include:

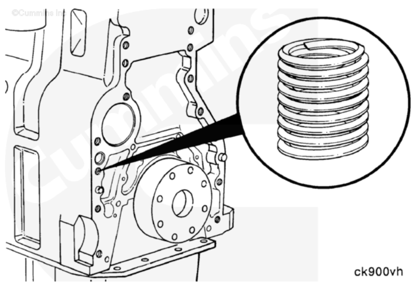
- Main bearing caps and bores
- Camshaft bores
- Cylinder bores
- Tappet bores
- Cylinder block combustion deck
- Oil pan mounting surface
- Lubricating oil pump mounting area
- Water pump mounting area
- Front and rear of block sealing surfaces
- Lubricating oil cooler cavity.

Make sure to inspect the main bearing caps and main bearing saddle areas for cracks, fretting and signs of discoloration.



If any cracks are found, the cylinder block **must** be replaced.

Inspect all threaded capscrew holes for damaged threads. Coiled thread inserts may be used to repair any damaged threads.



Service Tool threaded insert kits are available:

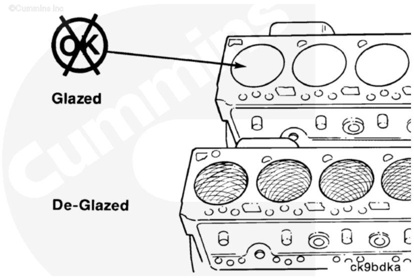
1. Part Number 3377905 for standard threads
2. Part Number 3377903 for metric threads.

Note : Coiled thread inserts must not be used to repair main bearing saddle threaded capscrew holes. If damaged, the block must be replaced.

Inspect the cylinder bores for glazing.

A surface without glaze will have a crosshatched appearance with the lines at 25- to 30-degree angles with the top of the cylinder bore.

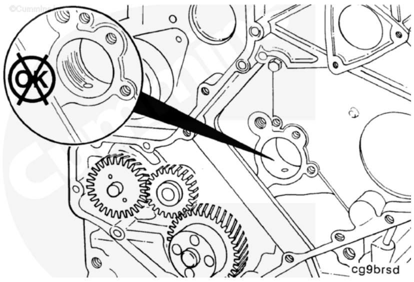
If deglazing is required, see the Deglazing section, located later in this section.



Inspect the camshaft bores for scoring, scuffing or excessive wear.

If damage to the camshaft bores is found and a camshaft bushing was **not** previously installed, machine the camshaft bores over size to install standard camshaft bushings. See the Measure step of this procedure for specifications.

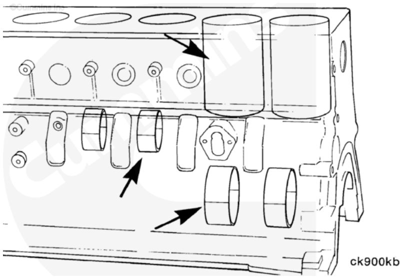
If the damage to the bore(s) is beyond machining, or if a camshaft bushing was previously installed, the block **must** be replaced. Oversize cam bushings are **not** available.



Measure

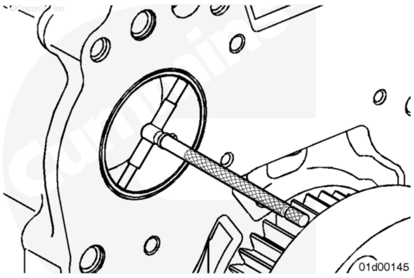
All measurements of the cylinder block **must** be made when the cylinder block is positioned on a flat surface with the main bearing caps installed, and torque plate installed.

If the cylinder block is mounted on the engine stand and/or the main bearing caps are **not** installed, the measurements can be incorrect because of distortion. (Cylinder bores, main bearing bores, camshaft bores, etc).



Inspect the camshaft bores without the camshaft bushing installed.

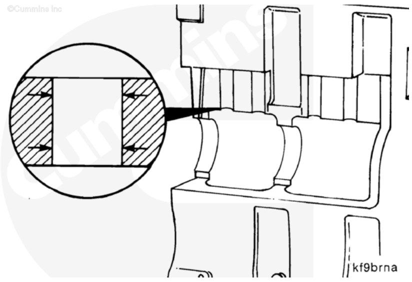
Use the following procedure for specifications. Refer to Procedure 001-010 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-010-tr.html)



Inspect the tappet bores for scores or excessive wear.

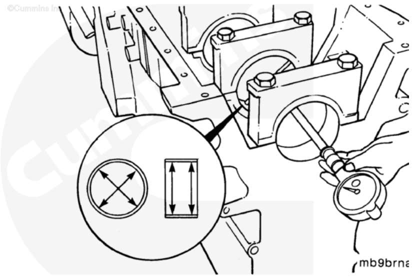
Measure the tappet bores.

Tappet Bore Diameter		
mm		in
16.000	MIN	0.630
16.055	MAX	0.632



Note : If the tappet bores are out of specification, the block must be replaced.

Install the main bearing caps without the main bearings. Use the following procedure for main bearing installation. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-006.html)



Tighten the main bearing cap capscrews.

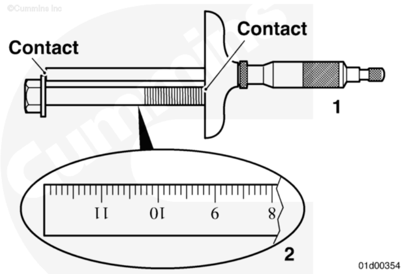
Torque Value: 176 n•m [130 ft-lb]

Measure the main bearing bore with the bearings removed.

Main Bearing Bore Diameter with Bearings Removed		
mm		in
87.983	MIN	3.4639
88.019	MAX	3.4653

Note : If the main bearing bore diameters are not within specification, check if the main bearing caps where installed in the proper location and orientation. If main caps are installed properly, the block must be replaced.

Main Bearing Capscrew Reuse Measurement



⚠ CAUTION ⚠

This step must be completed on B4.5 RGT engines. Failure to check the main bearing capscrew against reuse guidelines can result in severe engine damage.

To check if a main bearing capscrew can be reused, the length **must** be measured by performing the following:

For each main bearing capscrew that has been removed, measure the length from underneath the head of the capscrew to the tip of the capscrew, as illustrated, use one of two methods:

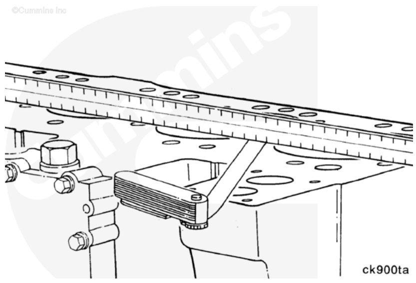
1. A depth micrometer (preferred method for accuracy)

2. A machinist's rule.

If the measurement is above the maximum specification, the main bearing capscrew **must** be replaced.

Main Bearing Underhead Capscrew Length		
mm		in
120.00	MAX	4.724

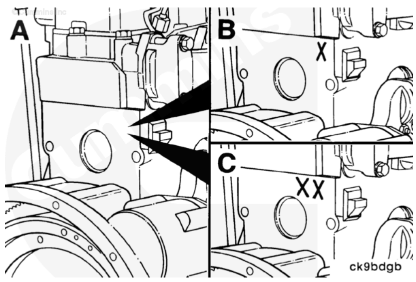
Measure the cylinder block's overall flatness.



Cylinder Block Flatness				
	mm		in	
End-to-End		0.076	MAX	0.003
Side-to-Side		0.051	MAX	0.002

Inspect for any localized dips or imperfections. If present, the deck **must** be resurfaced.

For B3.9, B4.5, and B5.9 series engines, oversize head gaskets are available for resurfacing of the cylinder head and cylinder block combustion decks.



oversize head gaskets **must** be used to make sure the correct piston to cylinder head clearance can be maintained.

Use a dial bore gauge accurate to within .001 mm [0.0001 in], calibrated with an appropriately sized master ring, to measure the cylinder bore in four places, 90 degrees apart, at the top and bottom of the piston travel area.

When measuring, deglazing, or boring, a cylinder block, make sure all of the main bearing caps, Cummins® cylinder head torque plate, and cylinder head gasket, are in place and properly torqued. Use the following procedures for proper torque values. Refer to Procedure 001-006 (/qs3/pubsys2/xml/en/procedures/40/40-001-006.html)in Section 1. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html) in Section 2.

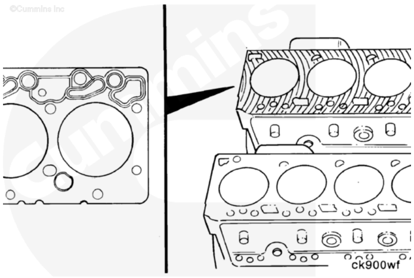
After machining, the cylinder block is stamped at the upper rear right corner surface of the cylinder block as follows:

A. Standard none

B. 0.25 mm [0.010 in] machined for the first over size gasket
   X
C. 0.50 mm [0.020 in] total for the second over size gasket
   XX.

For B4.5 RGT series engines, the combustion deck of the block can only be resurfaced if, after the resurface, the correct piston protrusion can be achieved.

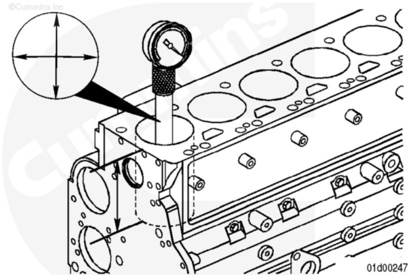
A specific head gasket with an increased thickness is **not** available for combustion deck resurfacing. If the combustion deck can **not** be resurfaced such that the correct piston protrusion can be achieved, the cylinder block **must** be replaced.



B3.9, B4.5, and B5.9 Series Engines

⚠ CAUTION ⚠

Do not measure the bore diameter within 50 mm (1.97 in) of the block combustion deck. Inaccurate measurements will result.



Use a dial bore gauge accurate to within .001 mm [0.0001 in], calibrated using an appropriately sized master ring, to measure the cylinder bore in four places, 90 degrees apart, at the top and bottom of the piston travel area.

When measuring, deglazing, or boring a cylinder block, make sure all of the main bearing caps, Cummins® cylinder head torque plate, and cylinder head gasket are in place and properly torqued. Use the following procedures for proper torque values:

Refer to Procedure 001-006 (/qs3/pubsys2/xml/en/procedures/40/40-001-006.html)in Section 1.

Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html) in Section 2.

Inspect the cylinder bores for damage or excessive wear.

Cylinder Bore Diameter - B3.9, B4.5, and B5.9 Series Engines Only (New Cylinder Block)		
mm		in
101.995	MIN	4.0155
102.045	MAX	4.0175

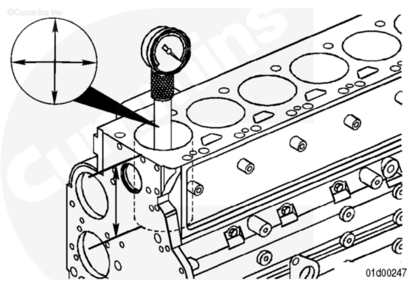
Cylinder Bore Diameter - B3.9, B4.5, and B5.9 Series Engines Only (Used Cylinder Block)		
mm		in
101.995	MIN	4.0155
102.070	MAX	4.0185

Out-of-Roundness		
mm		in
0.038	MAX	0.0015

Taper		
mm		in
0.076	MAX	0.003

Note : For B3.9, B4.5, and B5.9 Series Engines, the cylinders can be bored oversize twice for the use of oversize pistons and rings (0.5 mm [0.020 in] and 1 mm [0.040 in] over size). A repair sleeve can also be installed if the cylinder bore must be bored more than 1 mm [0.040 in] oversize. See the Overbore and/or Repair Sleeve section in this procedure.

B4.5 RGT Series Engines



⚠ CAUTION ⚠

Do not measure the bore diameter within 50 mm (1.97 in) of the block combustion deck. Inaccurate measurements will result.

Note : When measuring, deglazing, or boring a cylinder block, make sure all of the main bearing caps are in place and properly torqued. Refer to Procedure 001-006 (Bearings, Main) in Section 1 for proper torque values. (/qs3/pubsys2/xml/en/procedures/40/40-001-006.html)

Inspect the cylinder bores for damage or excessive wear.

Use a dial bore gauge to measure the cylinder bore in four places, 90 degrees apart, at the top and bottom of the piston travel area.

Cylinder Bore Diameter - B4.5 RGT Series Engines Only (New Cylinder Block)		
mm		in
106.990	MIN	4.2122
107.010	MAX	4.2130

Cylinder Bore Diameter - B4.5 RGT Series Engines Only (Used Cylinder Block)		
mm		in
106.990	MIN	4.2122
107.030	MAX	4.2138

Out-of-Roundness		
mm		in
0.038	MAX	0.0015

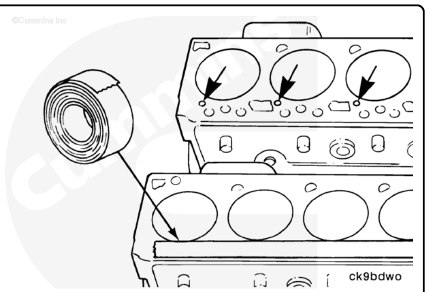
Taper		
mm		in
0.076	MAX	0.003

Note : For B4.5 RGT Series Engines, the cylinders can only be bored oversize once for the use of oversize pistons and rings (0.5 mm [0.020 in] oversize). A repair sleeve can be installed if the cylinder bore must be bored more than 1.5 mm [0.020 in] oversize. See the Overbore and/or Repair Sleeve section of this procedure.

Repair

⚠ CAUTION ⚠

Precautions must be taken to prevent debris from any reconditioning operation from entering the lubricating oil passages of the engine. Engine damage will result.

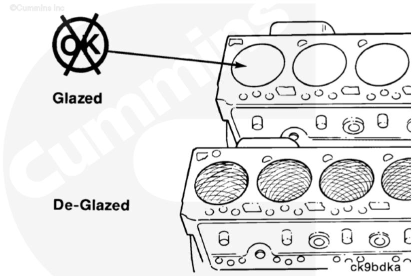


Prior to any reconditioning of the cylinder bores, make sure to cover the lubricating holes and tappet holes in the top of the cylinder block with waterproof tape.

Deglaze:

Deglazing gives the cylinder bore the correct surface finish required to seat the piston rings. Deglazing **must only** be performed if the cylinder bores are still in specification.

Note : New piston rings will not seat in glazed cylinder bores.



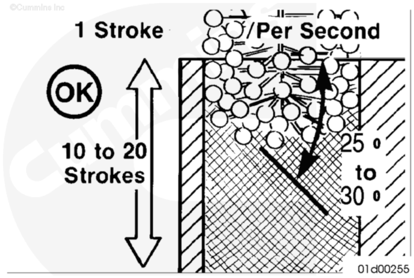
Use ball type hone and a rotational speed of 300 to 400 RPM with a stroke frequency of one stroke up and down per second. Make sure to use a good grade of honing oil or a mixture of equal parts SAE 30W engine oil and diesel fuel for a honing lubricant.

Note : Vertical strokes must be smooth, continuous passes along the full length of the cylinder bore.

Inspect the cylinder bore after 10 strokes.

Note : The crosshatch angle is a function of drill speed and how fast the hone is moved vertically. Moving too fast or too slow will give an incorrect crosshatch angle.

A correctly deglazed surface will have a crosshatched appearance with the lines at 25- to 30-degree angles with the top of the cylinder block.

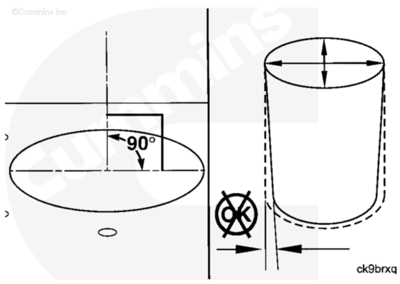


Overbore:

If the cylinder bore was found out of specification or damaged, the cylinder bore can be refinished for oversize pistons and piston rings.

Boring **must** be done by qualified personnel on a suitable boring machine. Care **must** be taken to make sure the cylinders are perpendicular to the combustion deck and within taper and out-of-round specifications for the cylinder bore.

Follow the boring machine manufacturer's recommendations for machine setup to achieve the best quality bore.

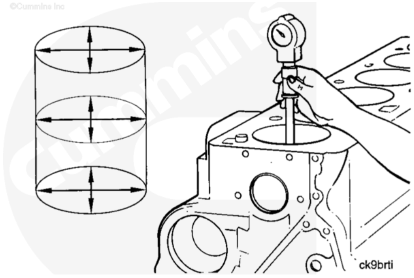


Note : The boring diameters given below are not the finished cylinder bore dimensions. The finished

cylinder bore diameter will be reached through the final honing operation.

The boring diameter dimensions are as follows:

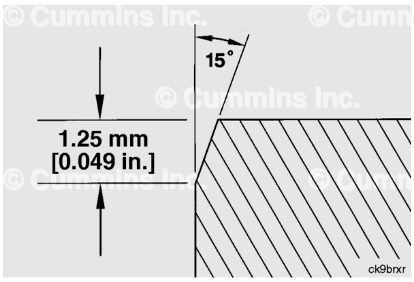
Note : Maximum cutting depth must be limited to 0.228 mm [0.009 in] per cut.



B3.9, B4.5, and B5.9 Series Engines Boring Diameter			
	mm		in
First Rebore	102.469	NOM	4.0342
Second Rebore	102.969	NOM	4.0539

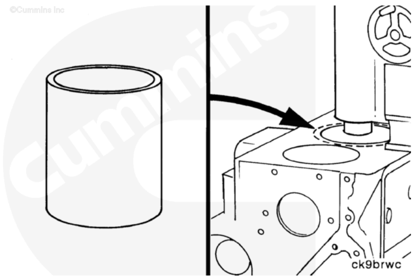
B4.5 RGT Series Engines Boring Diameter			
	mm		in
Rebore	107.45	NOM	4.2303

After boring, use a honing stone to break the edge of the bore to approximately 1.25 mm [0.049 in] at 15 degrees.



Repair Sleeve:

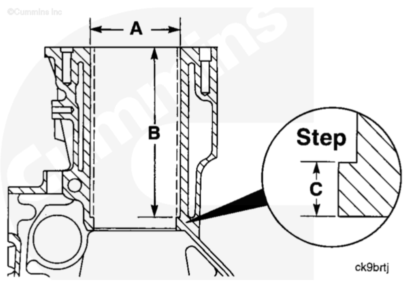
If more than 0.50 mm [0.020 in] in diameter oversize bore is required, the cylinder block **must** be rebored and a repair sleeve installed. The installation of a repair sleeve will allow for the use of standard size pistons and piston rings.



Note : Main bearing caps must be in place and properly torqued prior to any boring or measuring operations.

⚠ CAUTION ⚠

Boring must be done by qualified personnel on a suitable boring machine. Care must be taken to make sure that cylinders are perpendicular to the combustion deck and within taper and out-of-round specifications for the cylinder bore.



To prepare for repair sleeve installation, bore the cylinder(s) requiring a repair sleeve to:

Cylinder Bore Diameter (A)		
mm		in
109.700	MIN	4.3189
109.715	MAX	4.3195

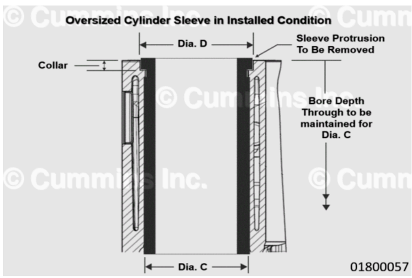
To a depth of:

Cylinder Bore Depth (B)		
mm		in
192.65	MAX	7.5846

This will result in a step at the bottom of the cylinder, approximately 6.35 mm [0.25 in] thick (C), against which the repair sleeve will sit.

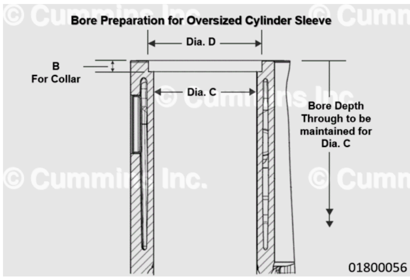
Oversized Cylinder Sleeves:

Oversized Cylinder Sleeves need a counter step at the top of the cylinder block for resting the collar of oversize sleeve.



⚠ CAUTION ⚠

Boring must be done by qualified personnel on a suitable boring machine. Care must be taken to make sure that cylinders are perpendicular to the combustion deck and within taper and out-of-round specifications for the cylinder bore.



Main bearing caps must be in place and properly torqued prior to any boring or measuring operations.

To prepare for oversize repair sleeve installation, bore the cylinder(s) through complete cylinder block bore depth and maintain bore diameters for respective size of Oversized Cylinder Sleeves:

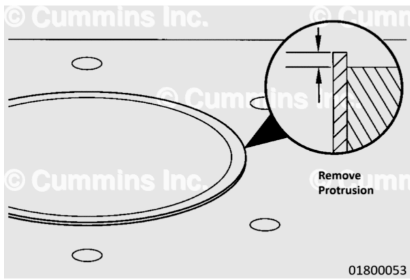
Oversized Cylinder Sleeves Dimensions

Part Number	Oversized Sleeves		Bore Diameter for Dimension C with tolerance			Bore Diameter for Dimension D with tolerance			HEIGHT B (mm) For Collar Seat		
	mm	in	mm		in	mm		in	mm		in
5628914	0.254	0.010	104.754	MIN	4.124	105.774	MIN	4.164	4.400	MIN	0.173
			104.769	MAX	4.124	105.794	MAX	4.165	4.600	MAX	0.181
5628915	0.508	0.020	105.008	MIN	4.134	106.028	MIN	4.174	4.400	MIN	0.173
			105.023	MAX	4.134	106.048	MAX	4.175	4.600	MAX	0.181

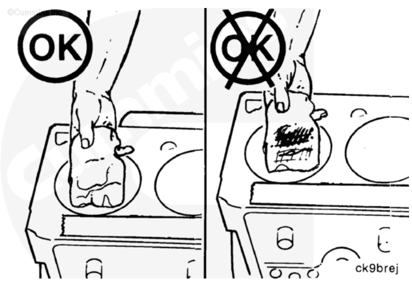
5628916	0.762	0.030	105.262	MIN	4.144	106.282	MIN	4.184	4.400	MIN	0.173
			105.277	MAX	4.144	106.302	MAX	4.185	4.600	MAX	0.181
5628917	1.016	0.040	105.516	MIN	4.154	106.536	MIN	4.194	4.400	MIN	0.173
			105.531	MAX	4.154	106.556	MAX	4.195	4.600	MAX	0.181

This will result in a step at the top of the cylinder, approximately 4.5 mm [0.177 in] in depth, against which the oversized repair sleeve will sit.

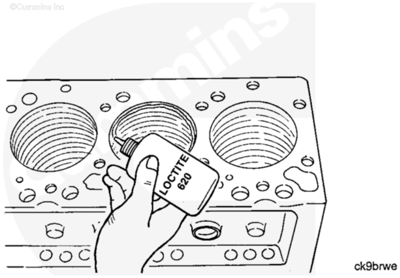
After installation, oversized repair sleeve will protrude out of top face by approximately 0.5 mm [0.020 in]. This sleeve protrusion needs to be removed by machining to flush it to the top face of the cylinder block surface.



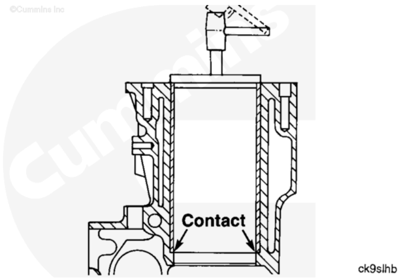
After boring, thoroughly clean the bore of all metal chips, debris and oil before installing the repair sleeve(s).



Apply a coat of Loctite 620 to the top of the cylinder bore, one at a time as the sleeve is installed.

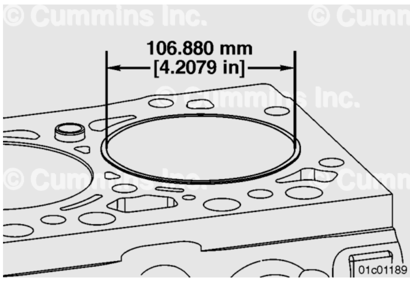


Use a sleeve driver, Part Number 3823230 for 102 mm [4.02 in], and Part Number 2892407 for 107 mm [4.21 in], to press or drive the repair sleeve into the cylinder bore until it contacts the step in the bottom of the bore.

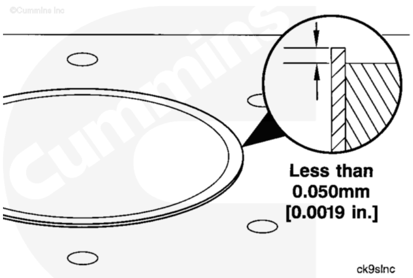


Standard Repair Sleeve for B4.5 engine series:

Bore the installed sleeve to 106.880 mm [4.2079 in].

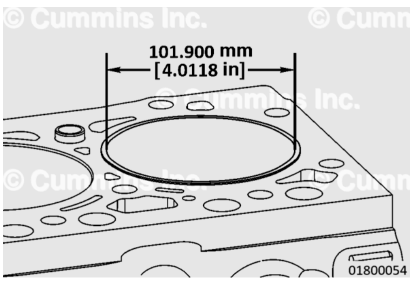


Machine the top of the sleeve to less than 0.050 mm [0.0019 in] protrusion above the combustion deck.



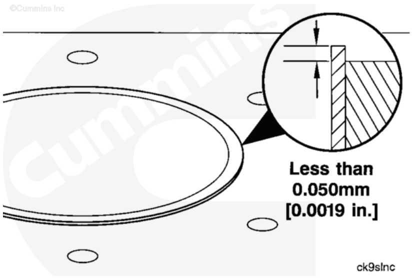
Standard Repair Sleeve for B3.9 and B5.9 engine series:

Bore the installed oversize sleeve ID to 101.900 mm [4.0118 in].



Machine the top of the sleeve to less than

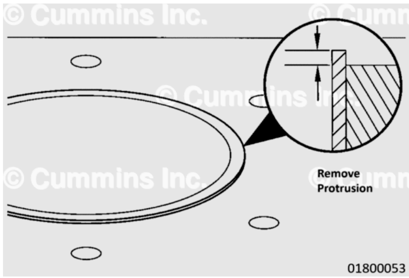
0.050 mm [0.0019 in] protrusion above the combustion deck



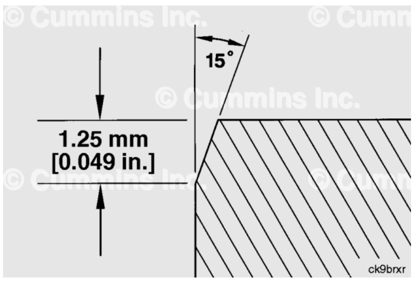
Oversized Cylinder Sleeve for B3.9 and B5.9 engine series:

Bore the installed oversize sleeve ID to 101.900 mm [4.0118 in].

After installation, oversize repair sleeve will protrude out of top face by approximately 0.5 mm [0.020 in]. This sleeve protrusion needs to be removed by machining to flush it to the top face of the cylinder block surface. Machine the top of the oversize sleeve to remove protrusion above the combustion deck.

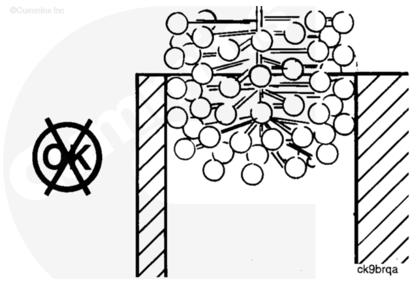


After boring, use a honing stone to break the edge of the bore to approximately 1.25 mm [0.049 in] at 15 degrees.

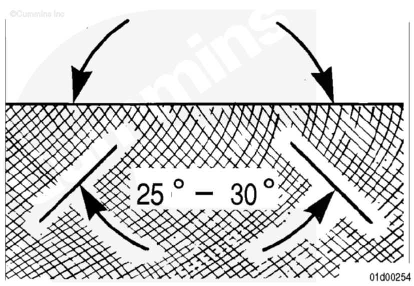


After boring a cylinder oversize or boring a repair sleeve, the cylinder requires a two stage honing process to finish the cylinder bores. It is recommended that quality equipment intended for honing engine cylinder bores be used.

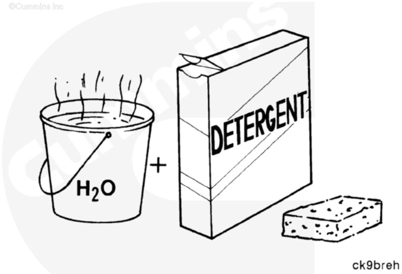
Note : Use of a ball-type hone is only recommended for refinishing cylinder walls that do need reboring and/or the installation of a repair sleeve.

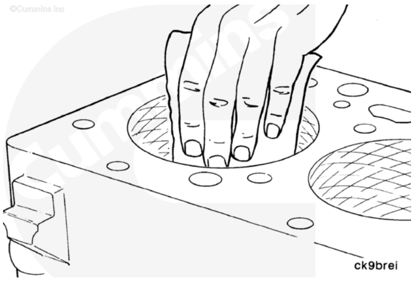


A correctly finished cylinder bore surface will have a crosshatched appearance with the lines at 25- to 30-degree angles with the top of the cylinder block.



After Deglazing/Finishing Honing, use a strong solution of hot water and laundry detergent to clean the cylinder bores.





⚠ WARNING ⚠

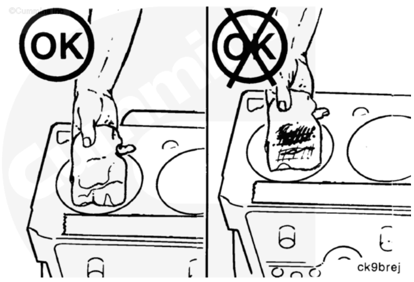
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean the cylinder bores immediately after deglazing/finish honing. Failure to do so can result in engine damage.

Rinse the cylinder bores until the detergent is removed.

Dry the cylinder block with compressed air.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

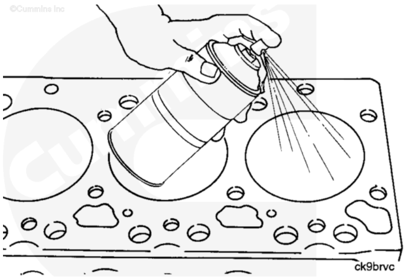
Be sure to remove the tape covering the tappet holes after the cleaning process is completed. Failure to do so can result in engine damage.

Check the cylinder bore cleanliness by wiping with a white, lint-free, lightly oiled cloth. If grit residue is still present, repeat the cleaning process until all residue is removed.

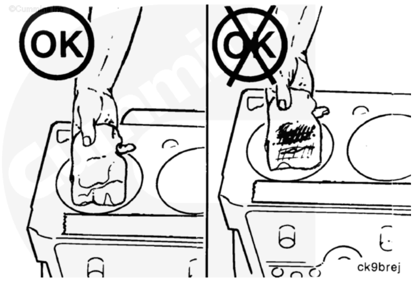
Wash the cylinder bores with solvent. Dry the cylinder block with compressed air.

If the cylinder block is **not** to be used right away, coat all machined surfaces with a rust preventative solvent.

Make sure to cover the cylinder block to prevent dust and debris from collecting on and in the cylinder block.

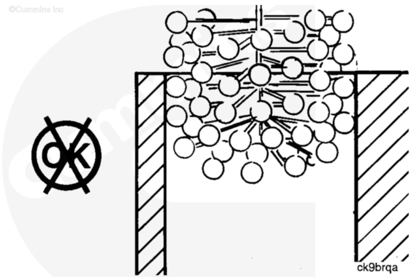


If replacing the cylinder block or using a previously stored cylinder block, make sure to clean any oil/rust preventative solvent from the cylinder bores, gasket sealing areas and main bearing bores prior to use.



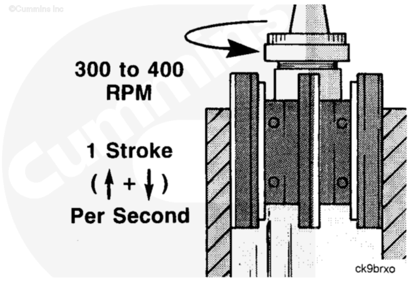
After boring a cylinder oversize, the cylinder requires a two stage honing process to finish the cylinder bores. It is recommended that quality equipment intended for honing engine cylinder bores be used.

Note : Use of a ball-type hone is only recommended for refinishing cylinder walls that do need reboring and/or the installation of a repair sleeve.



Honing Process for B3.9, B4.5, and B5.9 Engines

Use a honing rotational speed of 300 to 400 RPM with a stroke frequency of 1 stroke up and down per second. Make sure to use a good grade of honing oil. For the first stage honing, or rough honing, use a soft (fast cutting) 80 grit silicone carbide stone. Hone the cylinders to their final size during this stage of honing.



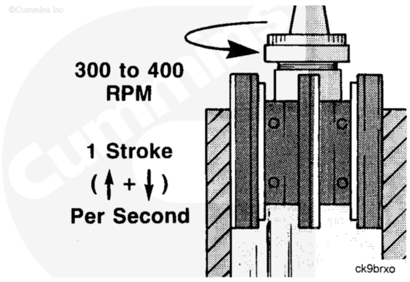
For the second stage honing, or finish honing, use a medium hardness 285 grit silicone carbide stone. Hone the cylinder(s) for 15 to 20 strokes to apply the appropriate crosshatch.

Honing Diameter Dimensions B3.9, B4.5, and B5.9 Series Engines			
	mm		in
Standard Bore/Repair Sleeve	102.000	MIN	4.0157
	102.040	MAX	4.0173
First Rebore	102.500	MIN	4.0354
	102.540	MAX	4.0370
Second Rebore	103.000	MIN	4.0551
	103.040	MAX	4.0567

Honing Diameter Dimensions B4.5 RGT and B6.7 Series Engines			
	mm		in
Standard Bore/Repair Sleeve	106.990	MIN	4.2122
	107.010	MAX	4.2130
Rebore	107.490	MIN	4.2319
	107.510	MAX	4.2327

Honing Process for Oversized Cylinder Sleeves

B3.9 and B5.9 Engines:



Use a honing rotational speed of 300 to 400 RPM with a stroke frequency of 1 stroke up and down per second. verify to use a good grade of honing oil. For the first stage honing, or rough honing, use 800 grit diamond tool.

Hone the cylinders to their final size during this stage of honing.

For the second stage honing, or finish honing, use 800 grit diamond. Hone the cylinder(s) for 10 to 20 strokes to apply the appropriate crosshatch.

Honing Diameter Dimensions for B3.9 and B5.9 Series Engines			
	mm		in

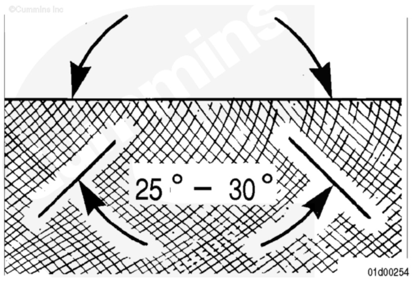
Oversized Cylinder Sleeve Bore	102.010	MIN	4.0161
	102.030	MAX	4.0169

Honing Process for B6.7 and B4.5 RGT Engines

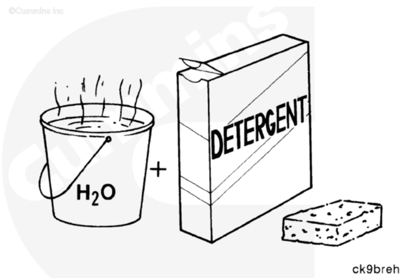
For the first stage honing, or rough honing, use a 160 grit diamond honing stone. Hone the cylinders to 106.9873 maximum.

For the second stage honing, or finish honing, use a 280 grit silicon carbide stone. Hone the cylinders to 106.990 minimum 107.000 maximum. Use a Plateau Honing Tools (PHT) brush for 10 to 12 strokes.

A correctly finished cylinder bore surface will have a crosshatched appearance with the lines at 25- to 30-degree angles with the top of the cylinder block.



After Deglazing/Finishing Honing, use a strong solution of hot water and laundry detergent to clean the cylinder bores.

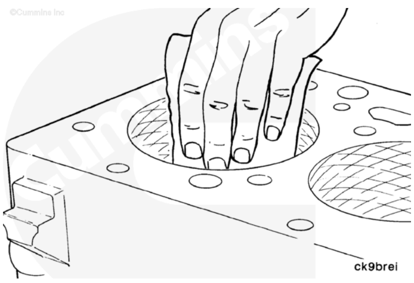


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean the cylinder bores immediately after deglazing/finish honing. Failure to do so can result in engine damage.

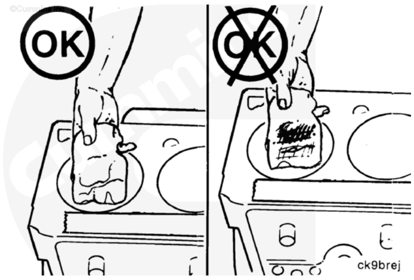


Rinse the cylinder bores until the detergent is removed.

Dry the cylinder block with compressed air.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

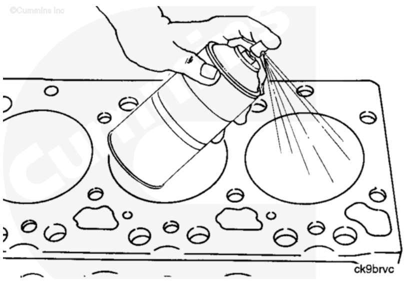
Be sure to remove the tape covering the tappet holes after the cleaning process is completed. Failure to do so can result in engine damage.

Check the cylinder bore cleanliness by wiping with a white, lint-free, lightly oiled cloth. If grit residue is still present, repeat the cleaning process until all residue is removed.

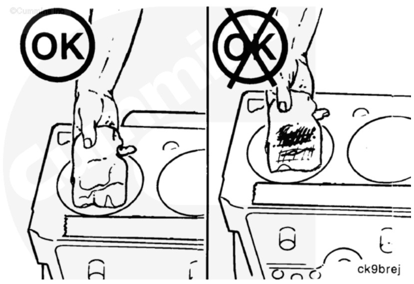
Wash the cylinder bores with solvent. Dry the cylinder block with compressed air.

If the cylinder block is **not** to be used right away, coat all machined surfaces with a rust preventative solvent.

Make sure to cover the cylinder block to prevent dust and debris from collecting on and in the cylinder block.



If replacing the cylinder block or using a previously stored cylinder block, make sure to clean any oil/rust preventative solvent from the cylinder bores, gasket sealing areas, and main bearing bores prior to use.



Finishing Steps

- Assemble the engine. See Section AS - Engine Assembly
- Remove the engine from the stand and install the engine. Refer to Procedure 000-002 (Engine Installation) in Section 0. (/qs3/pubsys2/xml/en/procedures/40/40-000-002.html)